


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
15-05-2024
JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

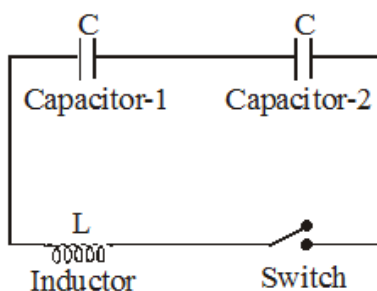
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

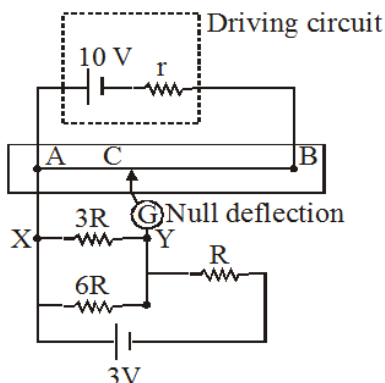
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider the circuit shown with respective specifications of elements marked in the figure. Capacitor-1 is charged such that charge on it is Q_0 and its left plate is positively charged. While capacitor-2 is uncharged. The switch is closed at $t = 0$.

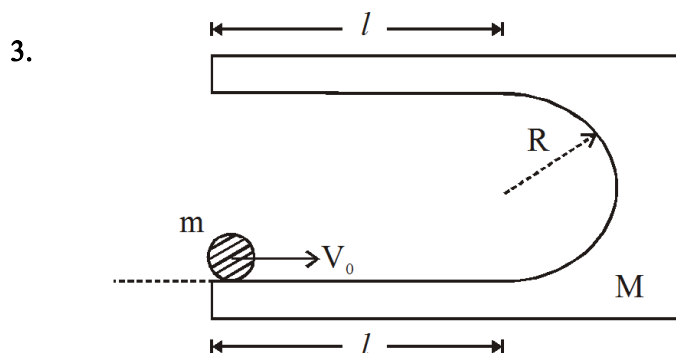


- (A) Frequency of oscillation of charge on left plate of capacitor-1 is $\frac{1}{\pi\sqrt{2LC}}$
- (B) Frequency of oscillation of charge on left plate of capacitor-1 is $\frac{1}{\pi}\sqrt{\frac{2}{LC}}$
- (C) Maximum current through the inductor is $\frac{Q_0}{\sqrt{2LC}}$
- (D) Maximum current through the inductor is $\frac{Q_0}{\sqrt{LC}}$

2. In the potentiometer circuit shown here, a potentiometer wire AB of length 100 cm and resistance $8\ \Omega$ is used. While measuring potential difference between points X and Y balancing length (AC) is obtained to be equal to 25 cm.



- (A) Internal resistance r (in Ω) of the driving cell is 2Ω .
 (B) Internal resistance r (in Ω) of the driving cell is 4Ω .
 (C) Potential difference between XY is 2 volt.
 (D) Potential difference between XY is 4 volt.

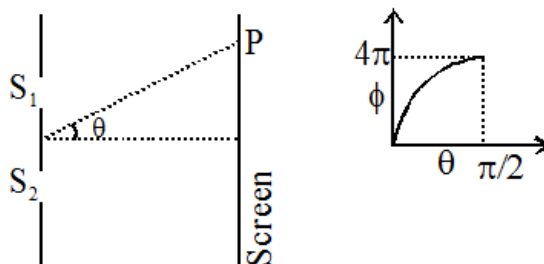


A particle of mass m moving with a velocity v_0 enters a stationary object of mass M (free to move) as shown in the diagram. All surfaces are frictionless and system shown in the diagram is lying in a horizontal plane.

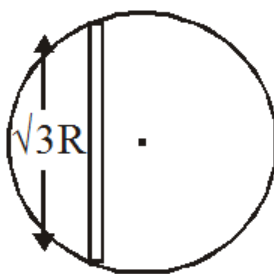
- (A) Time taken by particle of mass m to cross the object of mass M is $\frac{2\ell + \pi R}{V_0}$
 (B) Time taken by particle of mass m to cross the object of mass M is $\log e \frac{\ell + R}{V_0}$
 (C) Speed of particle (m) w.r.t block (M) will not change
 (D) Speed of particle (m) w.r.t block (M) will change
4. A coil with inductance L and resistance R is connected to an alternating source. What should be the capacity of a capacitor connected in series with the coil such that active power of circuit does not change.

- (A) $C = \frac{1}{\omega^2}$ (B) $C = \frac{1}{\omega^2 L}$ (C) $C = \frac{1}{2\omega^2 L}$ (D) $C = \frac{\omega^2}{L}$

5. In a YDSE setup, we plot the phase difference (ϕ) between both waves at point P on the screen against the angular position (θ) of point P on the screen. The graph is as shown below.



- (A) The distance $S_1S_2 = 2\lambda$
- (B) There are a total of 4 minima on the screen
- (C) The first maxima above the centre is at $\theta = \frac{\pi}{4}$
- (D) At $\theta = \frac{\pi}{6}$, intensity is maximum
6. A uniform rod of mass M and length $\sqrt{3}R$ is initially stationary and is in vertical position inside a frictionless, fixed hollow sphere of radius R . Now bar is released from rest from this position.



- (A) Find velocity of c.o.m when bar becomes horizontal $\sqrt{\frac{9R}{2}}$
- (B) Angular velocity of bar when it becomes horizontal is $\sqrt{\frac{2g}{R}}$
- (C) Normal reaction exerted by sphere at one end of rod when rod is horizontal is mg
- (D) Normal reaction exerted by sphere at one end of rod when rod is horizontal is $2mg$

SECTION-I (ii) : (Maximum Marks: 12)

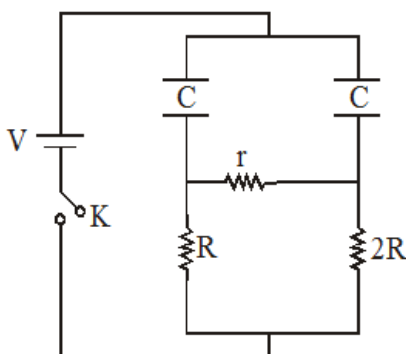
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Consider the situation shown in diagram. Both capacitors are having same capacitance C . Capacitors are initially uncharged. Switch K is closed at time $t = 0$ after a long time capacitors are fully charged. List- I represents value of r and List- II represents charge passing through r during complete charging process. Now match the followings :



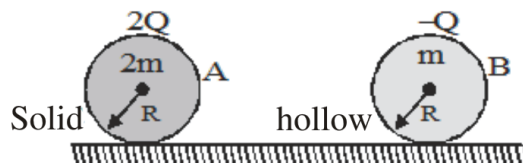
List- I		List- II	
(P)	$r = 0$	(1)	$\frac{CV}{6}$
(Q)	$r = R$	(2)	$\frac{CV}{5}$
(R)	$r = 2R$	(3)	$\frac{CV}{4}$
(S)	$r = 3R$	(4)	$\frac{CV}{3}$

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
 (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$
 (C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$
 (D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

8. Following is given the equation of a stationary wave (all in SI units) $y = (0.06) \sin(2\pi x) \cos(5\pi t)$

List- I		List- II (Only magnitude in SI units)	
(P)	Amplitude of constituent wave (in m)	(1)	0.06
(Q)	Position of node at $x = \dots$ (in m)	(2)	0.5
(R)	Position of antinode at $x = \dots$ (in m)	(3)	0.25
(S)	Amplitude at $x = 1/12$ m (in m)	(4)	0.03

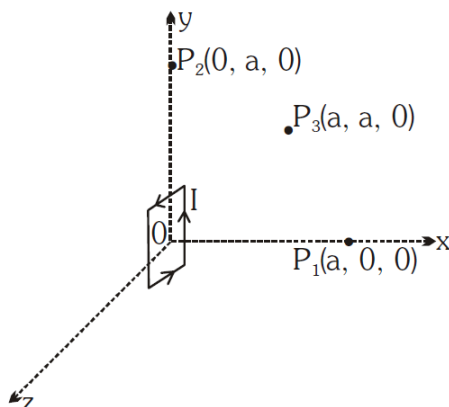
- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$
 (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$
 (C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$
 (D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$
9. Two charged di-electric spheres A (solid) and B (hollow) of same radius R and masses $2m$ and m respectively are kept on a horizontal rough surface. The two spheres A and B have charges $2Q$ and $-Q$ respectively and uniformly distributed. They are released from rest when their centres are separated by very large distance. Assuming there is no sliding.



List-I		List-II	
(P)	Kinetic energy of system just before A and B meet	(1)	$\frac{838kQ^2}{2345R}$
(Q)	Kinetic energy of A just before A and B meet	(2)	$\frac{kQ^2}{R}$
(R)	Kinetic energy of B just before A and B meet	(3)	$\frac{25kQ^2}{67R}$
(S)	Rotational kinetic energy of system just before A and B meet	(4)	$\frac{42kQ^2}{67R}$

- (A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$
 (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$
 (C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$
 (D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

10. A very small current carrying square loop (current I) of side ' L ' is placed in y - z plane with centre at origin of the coordinate system (shown in figure). In List-I the coordinate of the points are given & in List-II magnitude of strength of magnetic field is given. Then :-



List-I		List-II	
(P)	At point O (0, 0, 0)	(1)	$\frac{\mu_0 I L^2}{2\pi a^3}$
(Q)	At point P ₁ (a, 0, 0) (here $a \gg L$)	(2)	$\frac{2\sqrt{2}\mu_0 I}{\pi L}$
(R)	At point P ₂ (0, a, 0) (here $a \gg L$)	(3)	$\frac{\mu_0 \sqrt{5} I L^2}{16\pi a^3}$
(S)	At point P ₃ (a, a, 0) (here $a \gg L$)	(4)	$\frac{\mu_0 I L^2}{4\pi a^3}$

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
 (B) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$
 (C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$
 (D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

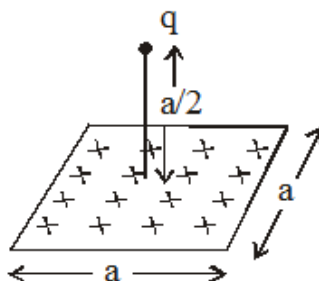
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

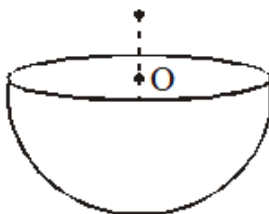
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A charge Q is uniformly distributed over a square plate of side a . Another point charge q is placed at a height $\frac{a}{2}$ from the centre of square plate as shown in diagram. Force acting on the point charge q is $\frac{nQq}{\epsilon_0 a^2}$. Find n .



2. A cell of emf 1.5 V and internal resistance 0.5Ω is connected to a nonlinear conductor in which current varies with voltage as $V^2 = I$. Calculate the current drawn from cell (in ampere).
3. Light falls on a prism of angle $\cos^{-1} \left(\frac{1}{8} \right)$. If the minimum value of refractive index of the prism-material, so that light entering one face may not exit through the other face
4. One mole of an ideal monatomic gas at temperature T_0 expands slowly according to the law $P/V = \text{constant}$. If the final temperature is $2T_0$, heat supplied to the gas is $\frac{RT_0}{x_0}$. Then find value of x ?
5. A hemispherical shell is fixed. The mass distribution of the shell is uniform. At a point on the symmetry axis, which is exactly at a distance of the radius measured from the centre of the hemisphere O , a small bead is released. The speed of the bead is v_0 when it reaches the centre O . The speed of the bead be when it reaches the shell is xv_0 . Find x (Neglect the effect of gravity.)



6. We stick a inextensible adhesive tape on a tabletop. We grab one end of it and pull it back horizontally and uniformly with a velocity of 6 cm/s. What is the velocity (in cm/s) of the middle of the moving part of the tape?



7. Two coils, 1 and 2, have a mutual inductance = M and resistance R each. A current flows in coil 1, which varies with time as : $i_1 = kt^2$, where k is a constant and 't' is time. The total charge that has flown through coil 2, between $t = 0$ and $t = T$ is $\left(\frac{\lambda}{-\lambda + 1} \right) \frac{kMT^2}{R}$. Find λ .
8. A small bead of mass m and positive charge $+q$ is mounted on a fixed, thin, smooth horizontal plastic ring of radius R . At $t = 0$, bead is stationary. Ring lies in x - y plane with centre at $(0, 0)$. Magnetic field $\vec{B} = \frac{E_0 t}{r} \hat{k}$ (where $r \neq 0$) is switched on, where t denotes time, r distance from z -axis and E_0 is a positive constant. Magnitude of normal reaction exerted by ring on bead is $qE_0 \lambda$. Find λ (Gravity is absent).

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

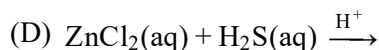
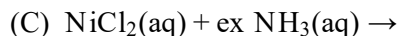
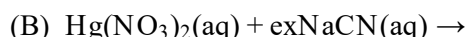
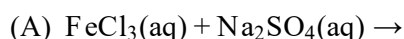
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

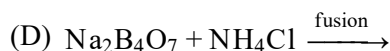
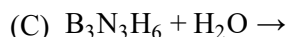
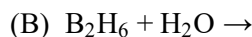
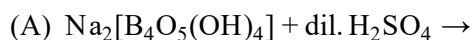
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

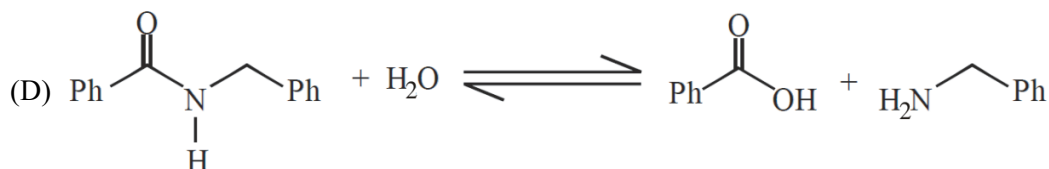
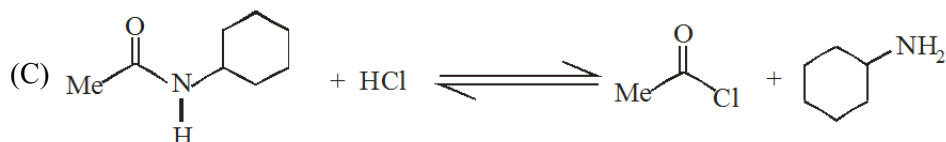
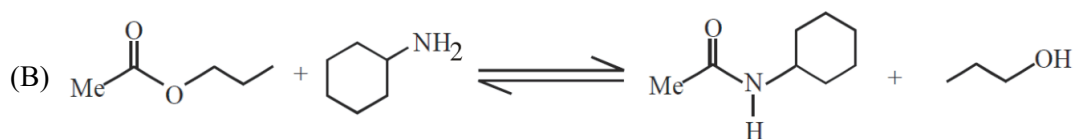
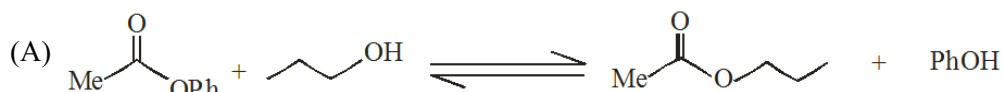
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1. How many of the following reactions will not result in precipitate formation ?
(ignore balancing)



2. Which of the following will result in formation of H_3BO_3 one of the products ?
(ignore balancing)



3. Which of the following statement(s) is(are) correct ?
- (A) Smoke and cloud both are aerosol with same dispersion medium.
- (B) Surface tension of lyophilic sols is less than the surface tension of dispersion medium while that of lyophobic sols is more than dispersion medium.
- (C) Lyophobic sols can be detected by ultra microscope
- (D) Colloidal sol shows low magnitude of colligative properties, as compared to true solutions.
4. If pH of an aqueous solution of 0.1M monoprotic acid HA is found to be 3 at T kelvin then
- (A) Degree of dissociation of the acid would be 1 % (approx.)
- (B) Osmotic pressure of the solution would be $0.011RT$
- (C) Theoretical molar mass of HA is approximately 1.01 times of its observed molar mass
- (D) K_b of $A_{(aq)}^-$ is 10^{-11} M (approx.) at 298K
5. Among the following, the reaction(s) that favor(s) the formation of the products at 25°C is/are



6. Among the following, the correct statements is/are:
- (A) The first pK_a of malonic acid is lower than the pK_a of acetic acid while its second pK_a is higher than the pK_a of acetic acid.
- (B) Picric acid shows positive test with $NaHCO_3$.
- (C) $CH(CN)_3$ is a stronger acid than HCN.
- (D) Amides are generally stronger acid than water due to resonance in their conjugate base.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

List-I (Complex)		List-II	
(P)	$[\text{Rh}(\text{PPh}_3)_3\text{Cl}]$	(1)	$d_{xy} = d_{yz} = d_{zx}$ (energy)
(Q)	$[\text{CoCl}_4]^{2-}$	(2)	High spin
(R)	$\text{Ni}(\text{CO})_4$	(3)	Low spin
(S)	$[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$	(4)	Diamagnetic

- (A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$ (B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$
 (C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$ (D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

8. Match the list

List-I		List-II	
(P)	Electrolysis of 100 L aqueous solution of CH_3COOK by passing 2 F of electricity	(1)	pH = 12.3 Anode = Ethane (g) + $\text{CO}_2(\text{g})$ Cathode = $\text{H}_2(\text{g})$
(Q)	Electrolysis of 10 L aqueous solution of HCOOK by passing 1 F of electricity.	(2)	pH = 13.0 Anode = $\text{H}_2(\text{g})$ + $\text{CO}_2(\text{g})$ Cathode = $\text{H}_2(\text{g})$
(R)	Electrolysis of 10 L aqueous solution of K_2SO_4 by passing 1 F of electricity	(3)	pH = 7.0 Anode = $\text{O}_2(\text{g})$ Cathode = $\text{H}_2(\text{g})$
(S)	Electrolysis of 10 L aqueous solution of CuF_2 by passing 1 F of electricity.	(4)	pH = 1.0 Anode = $\text{O}_2(\text{g})$ Cathode = $\text{Cu}(\text{s})$

- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$
 (B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$
 (C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$
 (D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

9. Match the list

List-I			List-II	
(P)	Ratio of rate of appearance (gm/min) of $\text{SO}_3(\text{g})$ and rate of disappearance (gm/min) of $\text{SO}_2(\text{g})$ for : $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$; would be		(1)	0
(Q)	Based on following the observations for a reaction $\text{A} \xrightarrow{\text{k}} \text{product}$ $[\text{A}]_0$ (in M) : 0.0677 0.136 0.272 $t_{1/2}$ (in min) : 2 4 8 the order of reaction would be		(2)	3
(R)	Time (min) taken by a reaction to decrease the concentration of the reactant(R) from 0.2 mole/litre to 0.025 mole/litre if its rate constant corresponding to rate of disappearance of R is 0.693 min^{-1} , would be		(3)	1.25
(S)	If emf at 25°C of the following Galvanic cell $\text{Pt} \text{H}_2(\text{g}) \underset{(\text{C}_1)}{\text{CH}_3\text{COOH}(\text{aq.})} \underset{(\text{C}_2)}{\text{HCOOH}(\text{aq.})} \text{H}_2(\text{g}) \text{Pt}$ is 0.0177 volt then ratio of $[\text{H}^+]$ in anodic compartment and cathodic compartment would be approximately		(4)	2

- (A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$ (B) $P \rightarrow 2; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
(C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 3; S \rightarrow 4$ (D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

10. Match the List-I with List-II

List-I		List-II	
(P)	Tyrosine	(1)	Gives effervescence with NaHCO_3
(Q)	O-Toluidine	(2)	Gives white precipitate with Bromine H_2O
(R)	Mesityl oxide	(3)	Gives alkali soluble product on reaction with Benzene sulphonyl chloride
(S)	Buna-N	(4)	Decolourizes cold KMnO_4
		(5)	Sodium fusion extract of the compound on boiling with FeSO_4 , followed by acidification gives prussian blue colour

- (A) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 1,2,3$
 (B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 1,4$
 (C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 4,5$
 (D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 2,3$

SECTION-II : (Maximum Marks: 24)

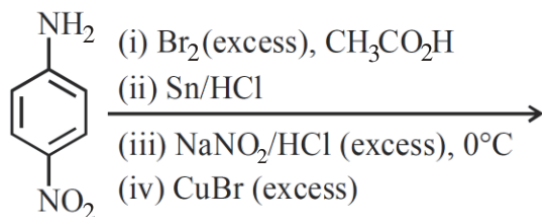
- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

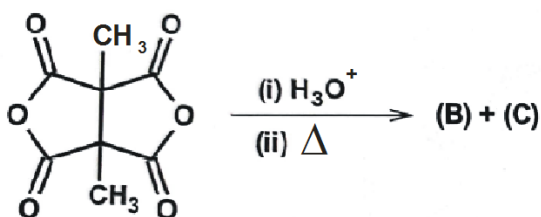
1. 0.1 mole of white phosphorus reacts with 0.6 mole SOCl_2 to form a pungent smelling gas(A) alongwith other products. Gas(A) is reacted with 0.2 mole KMnO_4 in acidic medium and the product mixture is treated with excess BaCl_2 solution. A white precipitate insoluble in dilute acids is formed. Let the weight of precipitate formed be x g. Find the value of $\frac{x}{3}$.
2. Consider following precipitates :
 CuS , ZnS , HgS , Ni(CN)_2 , AgCN , AgCl , CuCN , Pb(CN)_2 , FeS , MnS
Let,
x = number of precipitates soluble in aq.KCN
y = number of precipitates soluble in dil.HCl
Find $\frac{|x - y|}{2}$
3. Consider following minerals/ores :
Malachite, Calamine, Siderite, Kaolinite, Cryolite, Sphalerite, Beryl, Kernite
Let the number of types of elements present in all of the above be x. Find $\frac{x}{2}$
4. Total number of mixtures among the following that can form a maximum boiling azeotrope is/are -
(i) Carbon tetrachloride + Methanol (ii) Carbon disulphide + Acetone
(iii) Benzene + Toluene (iv) Benzoic acid + Aniline
(v) Chloroform + Ethanol (vi) Water + Ethanol
(vii) Bromobenzene + Chlorobenzene (viii) Bromoethane + Chloroethane
(ix) Suphuric acid + Water
5. In an aqueous solution, mole fraction of solute is 0.2 and its density is $d_1(\text{g/ml})$. It is diluted such that mole fraction of solute becomes 0.1 and density becomes $d_2 \text{ g/ml}$. The density of new solution is 20% less than initial solution and molarity of new solution is almost half of the original solution. If molar mass of the solute is 'X' gm/mol, then find X.

6. The molecular weight of the major product of the reaction is **X** and number of localized lone pair in final product are **Y**. Then $X \div Y$ is ?

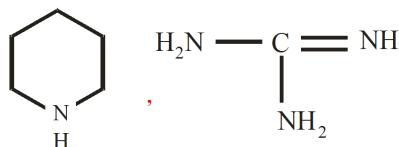


[Given : Atomic weight of H = 1, C = 12, N = 14 and Br = 80]

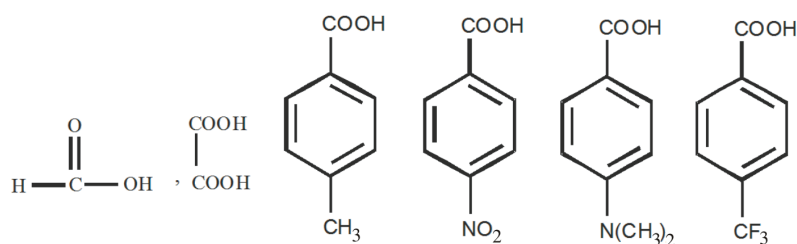
7. 1.1 mole of (A) was reacted according to below reaction. 'C' is the gas produced which turns lime water milky. The amount of 'C' formed in gram is **X**. The value of $X/14$ is
(Given At. Wt. of C, O, H are 12, 16, 1 gm, respectively)



8. **X** is the number of below compounds which are more basic than aniline?
o-toluidine, o-anisidine, m-anisidine, p-anisidine, p-chloroaniline, m-toluidine,



Y is the number of below compounds which are more acidic than benzoic acid



Picric acid, Hydrofluoric acid, Phosphoric acid (pK_{a1}), Hypochlorous acid

The value of (X/Y) is....

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. A letter lock consists of three rings marked with 15 different letters. If N denotes the number of ways in which it is possible to make unsuccessful attempts to open the lock, then
 - (A) N is divisible by 482
 - (B) N is product of 3 distinct prime numbers.
 - (C) N is product of 4 distinct prime numbers.
 - (D) The number of positive integer divisors of N is 8
 2. Twenty children are standing in a line outside a ticket window at Appu Ghar in New Delhi. Ten of these children have a one rupee coin each and the remaining 10 have a two rupee coin each. The entry ticket is priced at Re. 1. If all the arrangements of the 20 children are equally likely, the probability that the 10th child will be the first to wait for change is (assume that cashier has no change to begin with).
 - (A) 0
 - (B) $< \frac{1}{2}$
 - (C) $> 1/2$
 - (D) $1/2$

3. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is of magnitude $\sqrt{2/3}$ is
- (A) $2\hat{i} + 3\hat{j} - 3\hat{k}$
 (B) $2\hat{i} + 3\hat{j} + 3\hat{k}$
 (C) $-2\hat{i} - \hat{j} + 5\hat{k}$
 (D) $2\hat{i} + \hat{j} + 5\hat{k}$
4. Let a and c be two complex numbers. Then there is atleast one complex number z such that $|z - a| + |z + a| = 2|c|$ if
- (A) $|c| = |a|$
 (B) $|c| < |a|$
 (C) $|c| > |a|$
 (D) $2|c| < |a|$
5. Let F be the foot of perpendicular from $P(1, 2, -3)$ on the line $L = \frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$. Given $Q(x_1, y_1, z_1)$ and $R(x_2, y_2, z_2)$ are points at a distance of 3 units from F on line L . Let π_1 and π_2 be two planes passing through Q and R respectively and perpendicular to the line of intersection of the planes $9x - 7y + 6z + 48 = 0$ and $x + y - z = 7$. Then which of the following statement(s) is(are) correct?
- (A) The co-ordinates of F is $(1, 1, -1)$.
 (B) The value of $\sum_{i=1}^2 (x_i^2 + y_i^2 + z_i^2)$ equals 24.
 (C) The distance between planes π_1 and π_2 equals $\frac{88}{\sqrt{482}}$.
 (D) The normal vector of planes π_1 and π_2 is parallel to the vector $\hat{i} + 15\hat{j} + 16\hat{k}$.
6. Upon X - Y plane, the path defined by the equation $\frac{1}{x^m} + \frac{1}{y^m} + \frac{k}{(x+y)^n} = 0$, is
- (A) a parabola if $m = -\frac{1}{2}, k = -1, n = 0$
 (B) a hyperbola if $m = 1, k = -1, n = 0$
 (C) a parabola if $m = k = n = 1$
 (D) a pair of lines if $m = k = -1, n = 1$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.	List -I	List -II
(P)	Locus of centre of circles touching circles $x^2 + y^2 - 4x - 4y = 0$ internally and $x^2 + y^2 - 6x - 6y + 17 = 0$ externally	(1) straight line
(Q)	The locus of the point $(3h - 2, 3k)$, where (h, k) lies on the circle $x^2 + y^2 - 2x - 4y - 4 = 0$	(2) Circle
(R)	Locus of centre of circles touching two circle $x^2 + y^2 + 2x = 0$ and $x^2 + y^2 - 6x + 5 = 0$ externally is	(3) Ellipse
(S)	The extremities of a diagonal of a rectangle are $(0, 0)$ and $(4, 4)$. The locus of the extremities of the other diagonal is	(4) Hyperbola

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 2$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

8. Let A be a set containing n elements. P and Q are two subsets of A. The number of ways of Choosing P and Q so that

List -I	List -II
(P) $P \cup Q = A$	(1) 2^n
(Q) $P \cup Q = \phi$	(2) 3^n
(R) $P \cap Q$ is a singleton set	(3) $n \cdot 3^{n-1}$
(S) $P \cap Q = \phi$	(4) $4^n - 1$

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 2$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

9.

List -I		List -II	
(P)	Let $f(x)$ is a derivable function satisfying $f(x) = \int_0^x e^t \sin(x-t)dt$ and $g(x) = f''(x) - f(x)$, where $f''(x) = 2^{\text{nd}}$ order derivative with respect to x . Then the possible integers in the range of $g(x)$ is	(1)	1
(Q)	If the substitution $x = \tan^{-1}(t)$ transforms the differential equation $\frac{d^2y}{dt^2} + xy \frac{dy}{dx} + \sec^2 x = 0$ into a differential equation $(1+t^2)\frac{d^2y}{dt^2} + (2t + y \tan^{-1}(t)) \frac{dy}{dt} = k$ then k is equal to	(2)	0
(R)	If $a^2 + b^2 = 1$ then $a^3b - ab^3$ can be equal to	(3)	-1
(S)	system of equation $x - \lambda y - z = 0$ $\lambda x - y - z = 0$ $x + y - z = 0$ has a unique solution λ , can be	(4)	2

- (A) $P \rightarrow 1,2; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 2,4$
 (B) $P \rightarrow 1,2; Q \rightarrow 2,4; R \rightarrow 1; S \rightarrow 1,2$
 (C) $P \rightarrow 2,4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1,2$
 (D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1,2; S \rightarrow 2,4$

10.

List -I		List -II	
(P)	If the period of $\sin^2 2x + \cos^4 2x + 2$ is T; then $\frac{\pi}{T}$ is equal to	(1)	-2
(Q)	If $4^x - 2^{x+2} + 5 + \lambda - 1 - 3 = \sin y $, $x, y, \lambda \in \mathbb{R}$ then the possible values of λ is/are	(2)	2
(R)	Let $f(x) = \min \{x^2, 2\}$, then the possible integral values of 'a' for which $f(x) < a$ for atleast one real 'x' is/are	(3)	4
(S)	Let $f(x) = \begin{cases} 2x - 1 , & -2 \leq x \leq 1 \\ x^2 - 4, & 1 < x \leq 8 \end{cases}$, then possible integers in the domain of $y = f(f(x))$ is	(4)	6

(A) $P \rightarrow 3; Q \rightarrow 1, 3; R \rightarrow 1, 2, 3; S \rightarrow 2, 3, 4$

(B) $P \rightarrow 3; Q \rightarrow 1, 3; R \rightarrow 2, 3, 4; S \rightarrow 1, 2, 3$

(C) $P \rightarrow 1, 3; Q \rightarrow 1; R \rightarrow 3, 4; S \rightarrow 1, 2, 3$

(D) $P \rightarrow 2, 3, 4; Q \rightarrow 1, 2, 3; R \rightarrow 3; S \rightarrow 1, 3$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\int e^{\tan \theta} (1 - \tan \theta)^2 \cos^2 \theta d\theta = A e^{\tan \theta} \cos^B \theta + K$, then the value of $A + B$ is
2. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\sum_{k=0}^{r-1} \frac{1}{5^{n-1}} {}^nC_r {}^rC_k 3^k \right)$ is equal to
3. If z_1, z_2 are complex number such that $z_1^3 - 3z_1z_2^2 = 2$ and $3z_1^2z_2 - z_2^3 = 11$ then $|z_1^2 + z_2^2| =$
4. Let S be the focus of the parabola $y^2 = 8x$ and let PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is
5. If the line $\frac{x-1}{15} = \frac{y+1}{16} = \frac{z-1}{n}$ intersect the curve $6x^2 + 5y^2 = 1, z = 0$; then $10n^2 - 20n + k = 0$, where the value of $K - 2630$ is
6. Let A be a 2×3 matrix whereas B be a 3×2 matrix. If $\det(AB) = 4$, then the value of $\det(BA)$ is
7. If $y^2 = P(x)$ is a polynomial of degree 3, then $2 \frac{d}{dx} \left(y^3 \frac{d^2y}{dx^2} \right)$ equals $\frac{k}{5} P(x) \frac{d^3P(x)}{dx^3}$ then k is equal to
8. A sequence is obtained by deleting all perfect squares from set of natural numbers. The remainder when the 2003^{rd} term of new sequence is divided by 2048, is

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